



BSc Faculty of Computer Science and Engineering

Syllabus

Subject	Operating Systems			
	Type	Semester	ECTS	Code
	COMPULSORY (C)	2	5	40OSY152
Course Lecturer Course Assistant Course Tutor	MSc. Valdrin Haxhiu			
Aims and Objectives	This course is about the concepts, structure and mechanisms of operating systems. Its purpose is to present clearly the nature and characteristics of modern operating systems. It is not easy to describe operating systems because of the fact that operating systems are designed for many different computer systems including: embedded systems, smartphones, workstations, PC computers, mainframes and supercomputers. Understanding key principles of processes and threads, interprocess communication, the importance of processes, and thread synchronization, and the tools for their synchronization, CPU scheduling for process execution, the responsibilities for managing other resources like memories and input/output devices, file system and security aspects are the core objectives of this course.			
Learning Outcomes	On completion of this course, students will be able to: • have a thorough understanding of the basic structure and operation of an operating system as a software system that has a strong relation with the hardware of a computer system • understand the responsibilities of resource management in a computer system by the operating system • understand the importance of an operating system in providing a suitable execution environment for programs of a computer system user • understand the concepts of a process and a thread • understand the importance of process and thread synchronization • understand the problems and difficulties of process and thread synchronization • understand the proposed solutions to the problems of process and thread synchronization • understand the different ways of managing of I/O devices and standard I/O interfaces • understand the importance of protection and providing security for programs of a computer system user			
Course Content	Course Plan			Week
	Introduction to operating systems			1
	Operating system structures			2
	Processes			3
	Threads and concurrency			4
	CPU scheduling			5
	Process synchronization. Synchronization tools.			6
	Synchronization examples			7
	Deadlocks			8
	Memory management			9
	File system			10
	I/O management			11
	Protection			12

Teaching/Learning Methods	Teaching/Learning Activity		Weight (%)
	1.	Lectures and in-class activities	60%
	2.	Laboratory Exercises	40%
Assessment Methods	Assessment Activity	Number	Week
	1.	Final exam	100%
Course resources	Resources		Number
	1.	Lectures hall	1
	2.	Moodle platform	1
	3.	OS Simulator software	1
ECTS Workload	Activity	Weekly hrs	Total workload
	1.	Lectures + Exercises	24
Literature/References	<i>Operating System Concepts, 10th edition - Abraham Silberschatz, Peter Baer Galvin & Greg Gagne, Wiley (2018)</i>		
	<i>Modern Operating Systems, 5th edition - Andrew S. Tanenbaum & Herber Bos, Pearson (2022)</i>		
	<i>Operating Systems - Internals and design principles, 9th edition - William Stallings, Pearson (2018)</i>		
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